80 GHz Radar Level Sensor Development Kit

76–81 GHz FMCW radar for non-contact level and distance measurement, 20 m and 60 m versions

Development board. Each kit is ready to measure: the pre-flashed radar board with its antenna (20 m version: lens radome; 60 m version: horn and PTFE lens), a USB-to-TTL serial adapter with Dupont wires, the Kymopsis configuration GUI for Windows, UART protocol and wiring notes, and light engineering help with your setup. The level-measurement firmware is fixed and the board is not offered for reprogramming with other 77 GHz applications; the kit is intended for level measurement inside closed tanks and vessels. Firmware and GUI source code are not included. For custom firmware or modules in volume, use the B2B form at kymopsis.com/b2b/ or email support@kymopsis.com. GUI download, protocol notes and user guide: github.com/Kymopsis/radar-level-sensor-gui.

Overview

Two 80 GHz radar boards covering the full 76–81 GHz band, for non-contact distance and liquid or solid level measurement in tanks, silos, sumps, and open channels. The 20 m version has a PCB patch antenna and can take a lens radome for up to 30 m. The 60 m version feeds a waveguide horn with a PTFE lens for a narrow beam over long ranges. Both report the measured distance over UART and accept configuration commands for range, blind zone, offset, measurement interval, and false-echo learning.

Applications

- Liquid and solid level measurement in tanks and silos
- Sump, well, and open-channel level
- Non-contact distance measurement, drone height hold
- Level transmitter product development

Versions compared

	20 m version (PCB antenna)	60 m version (horn + lens)
Kit SKU	KYM-LVL80-DK-20	KYM-LVL80-DK-60
Frequency	76–81 GHz, FMCW	76–81 GHz, FMCW, 1 transmit / 1 receive
Transmit power	12 dBm	12 dBm
Antenna (in the kit)	PCB patch, 9 dBi, beam 100° × 80° (−3 dB, bare board); lens radome fitted	Waveguide horn with PTFE lens
Range	15 m typical, 20 m standard; up to 30 m with the lens radome	Up to 60 m
Accuracy (Kymopsis bench measurement)	±2 mm	±4 mm at any distance (long-range waveform, narrower sweep bandwidth)
Blind zone	6 cm	6 cm

Receiver noise figure	—	11 dB
Board size	Square board, 4 corner mounting holes	35.3 × 55.3 mm
Connectors	2.54 mm pin headers	2.0 mm dual-row headers: 2 × 8 (signals) and 2 × 4 (power)

Common specifications

Radar SoC	76–81 GHz single-chip radar with Arm Cortex-M3, 1 MB flash, 192 KB SRAM
Interfaces	UART0 (default, 115200 baud, distance and FFT output), UART1, SPI, I2C, SWD
Data output	Distance in metres (32-bit float), 1–5 reports per second; echo curve (512 FFT bins) on request; signal strength in dB
Configuration	Range, blind zone, sensor offset, measurement interval (≥ 100 ms, default 3500 ms), report rate, slave address 1–247, factory reset, false-echo learning (full, zone, remaining zone)
Supply (radar core)	3.1–3.5 V (3.3 V typical); 135 mA typical, 600 mA max; ripple ≤ 50 mV; source ≥ 1 A; a DC/DC supply should switch at ≥ 2 MHz
Analog input AIO1	Do not exceed 1.35 V (20 m version)
Operating temperature	−40 °C to 85 °C
Storage temperature	−40 °C to 85 °C

Accuracy values are Kympsis bench measurements of the two kits; the 60 m version holds ±4 mm at any distance because its long-range waveform uses a narrower sweep bandwidth. Other values are typical figures stated by the module maker. Performance depends on mounting, vessel, target, and scene.

Pin functions · 20 m version, pin headers (as printed on the board)

`3V3, GND`	Supply input (two 3.3 V inputs, J2 and J3)
`TX0, RX0`	UART0, distance output and commands
`TX1, RX1`	UART1
`TX2`	Auxiliary UART TX / GPIO
`DIO, SWCLK`	SWD debug and flashing
`CS, CLK, MOSI / CS1, SCLK1, MOSI1, MISO1`	SPI
`SCL1, SDA1`	I2C
`P12, P19, P28`	GPIO (P12 / P19 also boot configuration)
`AIO1`	Analog input, max 1.35 V

Pin functions · 60 m version, H1 signal header (2 × 8, 2.0 mm)

`1 TX0`	UART0 TX
`2 RX0`	UART0 RX
`3 TX1`	UART1 TX

4 RX1	UART1 RX
5 SPI1_CS	SPI chip select
6 SPI1_CLK	SPI clock
7 SPI1_MISO	SPI data out
8 SPI1_MOSI	SPI data in
9 BOOT1	Boot select; pull low before start-up to boot from internal flash
10 RST_N	Reset, active low
11 SCL1	I2C clock
12 SDA1	I2C data
13 SWDIO	SWD data
14 SWCLK	SWD clock
15 GND	Ground
16 AIO1	Analog input

Pin functions · 60 m version, H2 power header (2 × 4, 2.0 mm)

1, 3, 4	GND
2	5V_EN
5–8	5V_IN

Mechanical

Mounting	Above the target, as close to the vessel centre as possible, antenna facing the surface
Horn (60 m version)	Aluminium, Ø27 mm flange with 3 × M2 holes, 36.5 mm long, M14 × 1.5 thread for the PTFE lens
Boot configuration	BOOT0 high and BOOT1 low = boot from internal flash (both have internal pull-ups)
PC connection	UART0 at 115200 baud over Dupont jumper wires and a USB-to-TTL serial adapter set to 3.3 V logic (CH340 adapter and wires included in the kit)

Integration notes

- A narrow beam (lens or horn) keeps energy on the product surface and off ladders, agitators, inlets, and weld seams.
- Range depends on the lens, target size and reflectivity, and the scene; some variation in the usable range is normal.
- The radar is sensitive to supply noise: keep ripple below 50 mV and give it its own regulator.

Regulatory status. This board is sold as a development kit for engineers, developers, and integrators. It is not FCC certified, and this datasheet does not claim CE or any other certification. It is not an authorized transmitter for unrestricted installation or operation. A finished product that uses it needs its own equipment authorization; Kymopsis can scope that work through its B2B program but does not issue grants.